MATH9102: Fundamentals of Data Analysis

Unit 7: Regression Assumptions and Diagnostics
Checking a Model Before You Believe It

Today

1. **Controlling for a Third Variable** — partial correlation
2. **Three Models of Perceived Stress**
3. **What Regression Assumes** — and about what
4. **Normality of the Residuals**
5. **Unusual Cases** — outliers, leverage, influence
6. **Constant Variance**
7. **Collinearity**
8. **Models 2 and 3**
9. **Independence**
10. **Writing It Up**

Important Note

A diagnostic finds evidence of a problem, or finds none. Finding none is not proof that there is no problem — and the report says which.

Where This Week Sits

Unit 6 fitted models and read them. This unit asks whether to believe them.

What you have

- A fitted model, its coefficients and their intervals
- F, R^2, and nested models compared on the same rows
- Clustered pupils, and what they did to standard errors

What is new

- What "controlling for" means, as a correlation
- What each assumption is about, and how to check it
- Which cases the model leans on
- A diagnostic paragraph that claims only what it found

Checks come after fitting

The assumptions that matter most are about the residuals, and there are no residuals until there is a model.

Controlling for a Third Variable

Controlling for a Third Variable

Does feeling in control go with feeling less stressed — or do both just track how people like to present themselves?

The survey's three scales

- `tpcoiss`: Perceived Control of Internal States
- `tpstress`: Perceived Stress
- `tmarlow`: Marlowe–Crowne Social Desirability

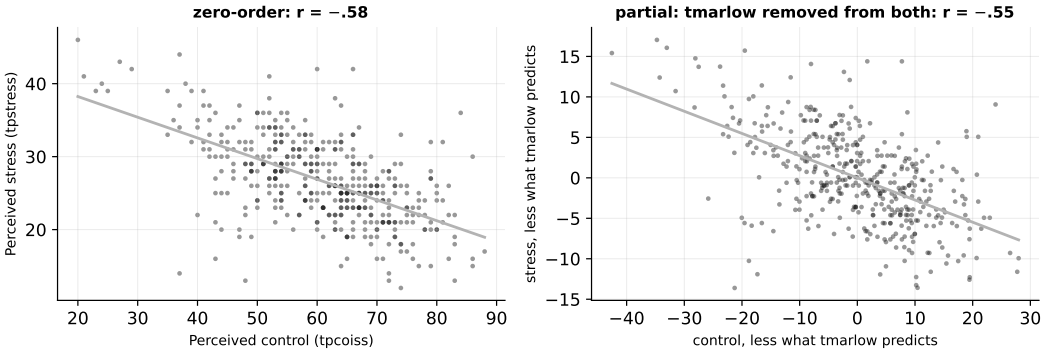
Socially desirable responding

Some people answer in the way that looks good. If they also claim calm and claim control, part of the correlation is that habit, not the two things being measured.

A partial correlation

Remove what the third variable predicts from *both* variables, then correlate what is left.

Zero-Order Against Partial



On the same 423 respondents: $r = -.58$ before, $-.55$ with social desirability removed from both. The relationship is barely touched: social desirability accounts for little of it.

Partial Is Not Semi-Partial

Two different questions, two different numbers.

Partial

The third variable removed from **both**. "Among people equally prone to look good, does control go with stress?"

Here: $-.55$

Semi-partial

The third variable removed from **one** of them. "Does control go with the part of stress that social desirability does not predict?"

Removed from stress only: $-.53$

Watch out

A published "partial correlation" of $-.53$ on these data is the semi-partial. The function was named spcor; the report said partial. Name the one you computed.

Reporting a Partial Correlation

```
import pingouin as pg

# Social desirability removed from both variables: a partial correlation.
result = pg.partial_corr(rows, x="tpcoiss", y="tpstress", covar="tmarlow")
m9.report_partial_correlation(rows, "tpcoiss", "tpstress", "tmarlow", result)
```

The paragraph carries

- $r(420) = -.55$, $p < .001$: the df are $n - 2 - q$ for q controlled variables
- The zero-order r on the same rows, so the change can be read
- That it describes association

It refuses a semi-partial

`m9.report_partial_correlation` recomputes the partial correlation and stops if the result it was given is anything else.

Three Models of Perceived Stress

How Many Cases?

Rules of thumb vary. Know where yours comes from.

Green (1991), for m predictors

- $N \geq 50 + 8m$ to test the whole model
- $N \geq 104 + m$ to test each coefficient

Both assume a medium-sized relationship.

For Model 1, with two predictors

- At least 66 for F, 106 for each b
- There are 423

A smaller effect needs more

These are floors for a medium effect, not targets. "Twenty cases per predictor" is quoted widely and sourced rarely: if you use it, treat it as a floor too, and say where it comes from.

Three Models of Perceived Stress

One outcome, `tpstress`; predictors added one at a time.

1. **Model 1:** perceived control, controlling for social desirability
2. **Model 2:** ... and optimism (`toptim`)
3. **Model 3:** ... and sex

Diagnosed, then compared

Model 1 is checked end to end. Models 2 and 3 are compared with it on the rows all three share.

Stand Up — 5 minutes

Five minutes. The models are named; nothing is fitted.

Food for thought

- Which way will control go with stress, once social desirability is in the model?
- Will optimism add much, given it overlaps with control?
- Out of four hundred people, how many will look unusual?

Say it out loud

Tell the person next to you, or jot it on paper. A prediction made before the output is a prediction. One made afterwards is a story.

Select Variables, Then Drop Rows

```
survey = m9.load_survey()

# The model's variables first, then the rows complete on those.
model_vars = ["tpstress", "tpcoiss", "tmarlow"]
print(m9.missingness(survey, model_vars).attrs["summary"])
rows = survey[model_vars].dropna()
```

Dropping rows across the whole file

- Keeps 333 of 439: 106 lost
- 79 of them for `smokenum`, which no model uses

Selecting the model's variables first

- Keeps 423, and loses only rows the model cannot use

Watch out

A row missing a variable you never use is not missing for your model.

Fitting Model 1

```
import statsmodels.formula.api as smf

model1 = smf.ols("tpstress ~ tpcoiss + tmarlow", data=rows).fit()
print(model1.summary())
```

- $F(2, 420) = 109.34$, $p < .001$; $R^2 = .342$
- Control: $b = -0.274$ per point, adjusted for social desirability
- Social desirability: $b = -0.191$, $p = .113$

Not yet believed

Every number on this slide rests on assumptions not yet checked. The rest of the session checks them.

What Regression Assumes

What Regression Assumes, and About What

Assumption	About	Checked with
Linearity	the mean of y at each x	residuals against fitted values
Normal errors	the **residuals**	Q–Q plot, then a test
Constant variance	the residuals' spread	residual plot, Breusch–Pagan
Independence	how the data were collected	the design — no plot can, unless the design names a grouping to plot by
No dominant cases	individual rows	leverage, Cook's D, DFBETAS
Predictors not redundant	the predictors	VIF

The outcome need not be normal

What is assumed normal is the error around the line. A skewed outcome can have normal residuals, and a normal one can fail. Categorical predictors are fine: they enter as dummies.

Non-Zero Variance

A variable that does not vary cannot predict, or be predicted.

Variances on Model 1's rows

- Perceived stress: 34.24
- Perceived control: 144.04
- Social desirability: 4.06

A check, not a ritual

It fails when a filter leaves one value — every respondent in a subgroup gave the same answer. Descriptives catch it before the model does.

Residuals Are What We Check

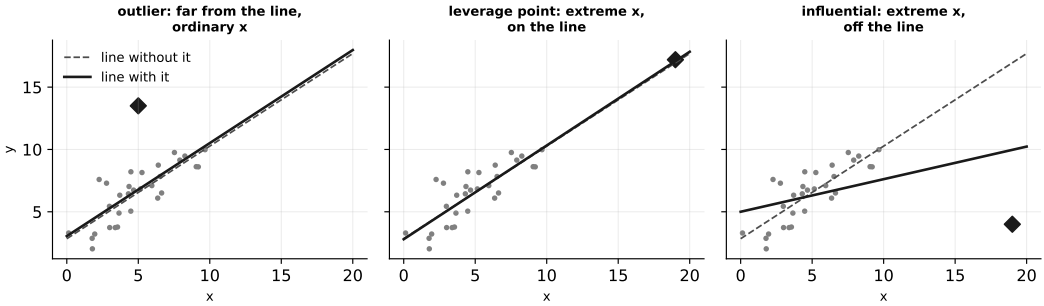
A residual is the part of a score the model did not predict.

- $e_i = y_i - \hat{y}_i$, one per case
- Least squares makes the sum of their squares as small as it can
- Their pattern shows what the line missed

Three things to look for

- A shape: a curve the line cut through
- A spread that changes along the line
- A few cases far from everyone else

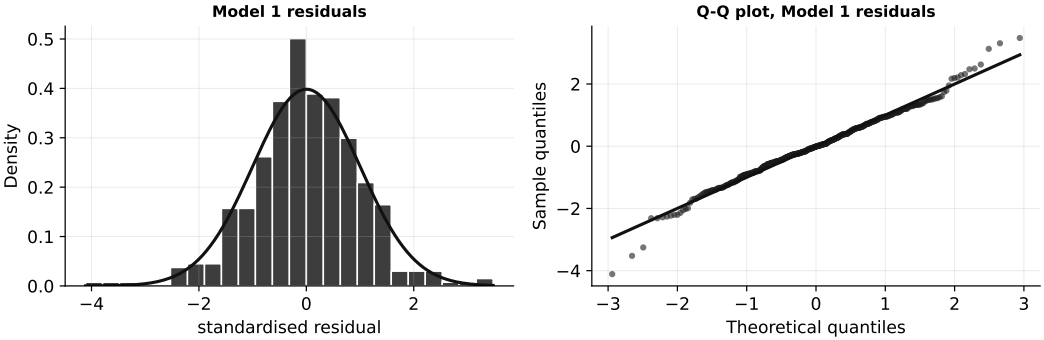
Outliers, Leverage and Influence Are Different Things



An illustration, thirty seeded points. An **outlier** is far from the line in y; a **leverage point** is far from the others in x. A case is **influential** when removing it moves the line — typically both at once.

Normality of the Residuals

Normality of the Residuals, Visually First



Model 1's residuals, and only Model 1's: a histogram with the normal curve beside a Q–Q plot. Close to the line through the middle; both tails stray a little, the lowest residual furthest.

What the Test Adds

Shapiro–Wilk on the residuals: $W = .988$, $p = .001$.

At this sample size

With 423 residuals the test detects departures far too small to matter for the model. Significant here is not a verdict; the Q–Q plot is the evidence.

And the other way round

In a small sample a non-significant test is weak evidence of anything. It found no problem; it did not show there is none.

With hundreds of cases the coefficients' intervals hold up under moderate non-normality. A handful of extreme cases can break them — the next section — and so can a changing spread or cases that are not independent, the sections after.

Unusual Cases

What a Standardised Residual Is

Each residual divided by its own standard error.

The definition

$$r_i = \frac{e_i}{s\sqrt{1 - h_{ii}}}$$

s: the residual standard error. h_{ii}: case i's leverage. A case far out in x pulls the line towards itself, so its residual is smaller than it should be; the denominator corrects for that.

Not $e/\mathrm{sd}(e)$

Dividing every residual by one standard deviation ignores leverage. Here the two ranges differ only slightly — -4.11 to 3.48 against -4.10 to 3.48 — but with a high-leverage case they can differ a lot.

Large Residuals

Two conventions, both from the normal curve.

- About 5% of residuals beyond ± 1.96 is expected
- Beyond ± 3.29 — a two-sided 1-in-1,000 — is worth a look (Tabachnick and Fidell)

Model 1

- Range -4.11 to 3.48
- 5.9% beyond ± 1.96; 4 beyond ± 3.29

More than normality predicts

A normal distribution would put fewer than one of 423 cases beyond ± 3.29. The tails are heavier than normal. The question is whether any of these cases changes the model.

The Bonferroni Outlier Test

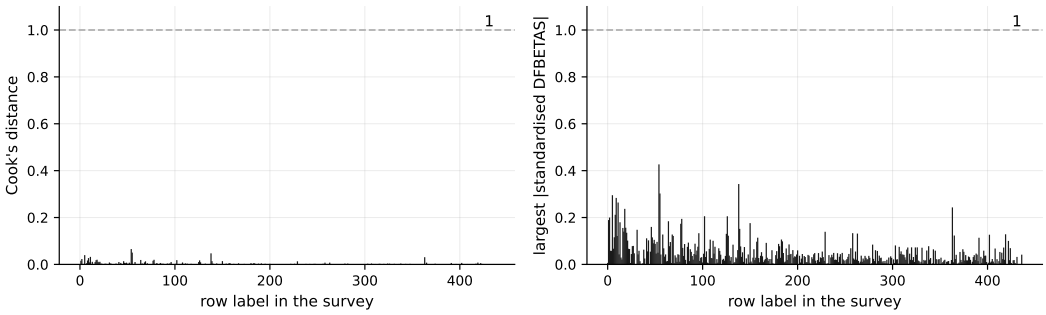
```
# Studentised residuals, with Bonferroni-adjusted p-values,
# the most extreme first. The index is the row label in the
# data, so the case can be looked up.
tests = model1.outlier_test()
tests.sort_values("student_resid", key=abs, ascending=False).head(3)
```

- Studentised, not standardised: s comes from the model fitted without the case, so it cannot hide its own residual
- The largest is tested, with its p multiplied by the number of cases — because it was picked as the largest
- Row 54: $t = -4.19$, Bonferroni $p = .014$

Which case?

Name the case by its row in your data. A position in a filtered vector refers to nothing anyone can look up.

Cook's Distance



Cook's D**:** how far all the fitted values move when a case is left out. Above 1 is the usual alarm; $4/n$ is a stricter screen, and conventions differ. **DFBETAS:** how far each coefficient moves, in its own standard errors; above 1 is the usual convention. Largest here: 0.066 and 0.43.

Standardised DFBETAS

```
# The cases the model leans on most: Cook's distance, leverage,
# standardised residuals and the largest standardised DFBETAS.
m9.check_influence(model1, top=5)
```

- One value per case per coefficient: which coefficient a case leans on, not just whether it leans
- Leverage above $2k/n = .014$, for k coefficients counting the intercept: 26 cases — far in x, harmless unless they also move the line

Standardised, and a convention

1 is a convention, and it belongs to the *standardised* DFBETAS: in a coefficient's own units it means nothing. Belsley, Kuh and Welsch's size-adjusted $2/\sqrt{n} = .097$ lists 51 cases here: a list to look at, not an alarm.

What To Do With an Outlier

```
# Fit without the case, and compare. Never delete it silently.
worst = model1.outlier_test()["student_resid"].abs().idxmax()
without = smf.ols("tpstress ~ tpcoiss + tmarlow",
                  data=rows.drop(index=worst)).fit()
pd.DataFrame({"with": model1.params, "without": without.params})
```

Run it twice

- With row 54: $b = -0.274$, $R^2 = .342$
- Without it: $b = -0.282$, $R^2 = .362$
- Same direction, same conclusion: keep it, and report both

Never delete silently

Check it for an entry error. A real case stays, unless it is outside the population the question is about — decided before fitting, stated, and counted. Deleting cases because the model fits them badly flatters the fit.

Break — 30 minutes: Model 1's Residuals and Cases

20 minutes on the analysis, then 10 for tea.

In `notebook.ipynb`

- **Parts 1 and 2**, as far as the outlier run twice

Then in `exercise.ipynb`

- Questions 1 and 2 — count before you model, and recode `stressmo`

Think about

Which of your stand-up predictions held? Look at row 54 in the data: is there anything about it that says "error"?

Then stop

The last ten minutes are tea, not catch-up.

Where You Should Be: Residuals and Cases

You should now have

- The partial correlation, reported beside the zero-order one
- Model 1 fitted on its own 423 rows
- Its residuals read from the plots first, then the test
- The one case the outlier test names, run with and without

The two that catch people

Reading a significant Shapiro–Wilk at four hundred cases as a verdict. And deleting a case because it is unusual, rather than because it is wrong.

Constant Variance

Constant Variance

The residuals should spread about as widely everywhere along the line.

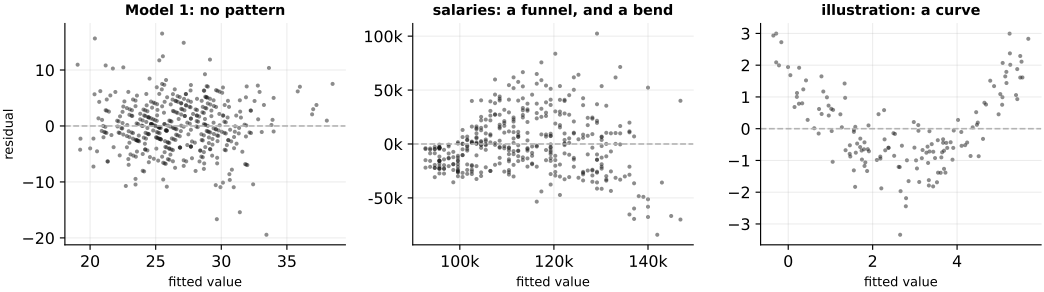
What goes wrong if they do not

The coefficients stay unbiased. Their **standard errors** are wrong — so are the intervals and *p*-values built on them. Often too small, so too many "significant" findings.

Not "bias"

Heteroscedasticity does not bias least-squares coefficients, and it does not make a model "invalid". It makes the reported standard errors wrong, and those can be repaired.

Reading Residuals Against Fitted Values



No pattern: what you hope for. **A funnel:** spread grows with the fitted value — professors' salaries against years since PhD, which bend down at the top as well. **A curve:** the line cut through a bend (an illustration). Act on a clear pattern, not on every wobble.

Which Breusch–Pagan?

```
from statsmodels.stats.diagnostic import het_breuschpagan

# Koenker's version (statsmodels' default): LM statistic and its p-value.
lm, lm_p, f, f_p = het_breuschpagan(model1.resid, model1.model.exog)
m9.check_homoscedasticity(model1)
```

Model 1, two versions

- Koenker's, on the predictors: $\chi^2(2) = 3.33$, $p = .189$
- The original, on the fitted values: $\chi^2(1) = 3.24$, $p = .072$

This module uses Koenker's

It does not assume normal residuals, and these are not quite normal. R's `performance` package runs the original, so its p-value differs. Name the version you report.

Neither finds evidence of changing variance — which is not proof of constant variance. The plot is read as well.

Remedies: Robust Standard Errors, or Transform the Outcome

```
salaries = m9.load_salaries()
formula = "salary ~ yrs_since_phd"

ols = smf.ols(formula, data=salaries).fit()
hc3 = smf.ols(formula, data=salaries).fit(cov_type="HC3")
pd.DataFrame({"b": ols.params, "SE, OLS": ols.bse, "SE, HC3": hc3.bse})
```

Salaries: Breusch–Pagan 50.8, $p < .001$

- Slope 985 dollars a year either way
- Standard error 107 by OLS, 128 robust (HC3)

Robust standard errors first

They repair the reported standard errors and leave the question unchanged. A transformation changes the question.

Interpreting a Transformed Outcome

Transform the outcome, and every coefficient changes its units.

- Raw salary: dollars per year since PhD
- log(salary): a proportion — each year goes with about 0.86% more
- $\sqrt{y}$: square-root units, which nobody can picture

It may not even work

Logging salary does not close the funnel: Breusch–Pagan is still 38.2, and still significant. Check the residuals of the model you report, not the one you meant to fit.

Collinearity

Collinearity

Two predictors carrying largely the same information.

What it does

The model cannot tell which of them deserves the credit, so their standard errors grow. Coefficients can swing or change sign between samples.

What it does not do

It does not bias the coefficients, and it does not make the model "invalid": predictions from the whole model are fine, within the combinations of predictor values in the data. It makes individual coefficients imprecise.

A correlation matrix of the predictors is a first look; it misses a predictor that is a combination of several others. The VIF does not.

VIF and Tolerance

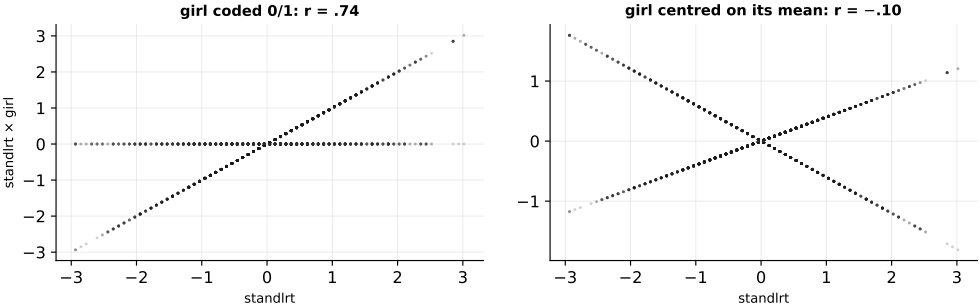
```
# One row per term: VIF, tolerance, and the generalised VIF
# for a categorical block.
m9.check_collinearity(model1)
```

- VIF: how many times larger a coefficient's variance is than if its predictor were unrelated to the others
- Tolerance $= 1/\text{VIF}$: one rule, not two
- VIF above 5 investigate, above 10 serious — conventions

The three models

- Largest VIF: 1.10, 1.47, 1.51
- Adding optimism raises it, a little

A Product Term Is Collinear by Construction



Unit 6's interaction model: `standlrt` has VIF 2.23, because for every boy the product is zero and for every girl it equals `standlrt`. Centre `girl` on its mean and it falls to 1.01. Same fit, same interaction test: this collinearity is structural, and harmless. Centring does change what the `standlrt` coefficient means — the slope at the mean of `girl`, not boys' slope — and it barely moves here only because the interaction is near zero.

Categories: One Block, One Number

A categorical predictor's dummies are checked together.

Youth Cohort school type, six categories

- Alphabetical reference: 116 students
- A VIF per dummy: up to 28.9 — "serious"
- One generalised VIF for the block: 1.02 on the VIF scale ($\text{GVIF}^{1/(2\,df)} = 1.01$) — whichever category is the reference

The generalised VIF

Fox and Monette (1992).
`m9.check_collinearity` reports one row per term, on the VIF scale, so 5 and 10 read the same way. The per-dummy number measured the small reference group, not collinearity.

What To Do About Collinearity

Only if it is high, and only if you need the separate coefficients.

- **Keep one** of the overlapping predictors, as the measure of the shared idea
- **Combine them** into one score — if the combination makes sense on its own terms
- **Estimate what they share**: a component or factor score, which is Unit 8
- **Leave them**, if the question is the whole model's prediction rather than any one coefficient

Models 2 and 3

Model 2: Adding Optimism

```
# The rows all three models share, then each model on them.
common = survey[["tpstress", "tpcoiss", "tmarlow", "toptim", "sex"]].dropna()
m1 = smf.ols("tpstress ~ tpcoiss + tmarlow", data=common).fit()
m2 = smf.ols("tpstress ~ tpcoiss + tmarlow + toptim", data=common).fit()
m9.report_model_comparison(m1, m2)
```

On the 422 rows all three models share

- Optimism adds .039 to R^2: $F(1, 418) = 26.18$, $p < .001$
- Largest VIF 1.47: optimism and control overlap, not dangerously

Same rows, every model

One row short of Model 1's own sample: one respondent has no optimism score. Compare on the shared rows, or not at all.

Model 3: Adding Sex

Does sex add anything, at the same control, desirability and optimism?

- R^2 rises by .013: $F(1, 417) = 9.02$, $p = .003$
- Largest VIF 1.51
- Diagnosed the same way as Model 1 — in the notebook

An extra predictor, not a moderator

Model 3 shifts the line for one sex. Whether sex *changes the effect* of control is an interaction, and Model 3 has none. "Moderates" needs the product term.

Stand Up — 5 minutes

Five minutes. One assumption is left, and no plot can check it — unless the design says what to group by.

Food for thought

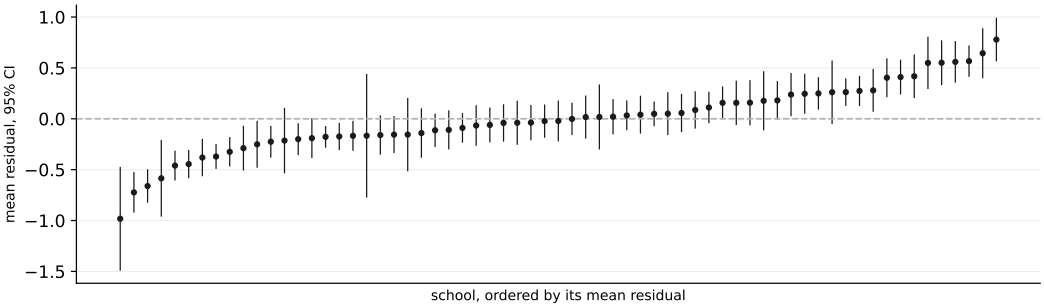
- The survey asked each person once. What could make two answers not independent?
- Unit 6's pupils sat in schools. What did that do?
- Could a test on the residuals have found it?

Say it out loud

Tell the person next to you, or jot it on paper — not on a screen.

Independence

Independence: Pupils in Schools Again



Unit 6's Model 1 residuals, averaged by school. Independent residuals would leave nearly every school's interval across zero; 31 of 65 miss it. The residuals' ICC is .128: for a mean, a design effect of 8.9. For the reading slope, cluster-robust standard errors are 1.84 times the ordinary ones.

Independence Is a Design Question

Ask how the data were collected, not what the residuals look like.

Not independent

- People in the same school, ward or household
- The same person measured twice
- Observations in time order, each like the last

The survey

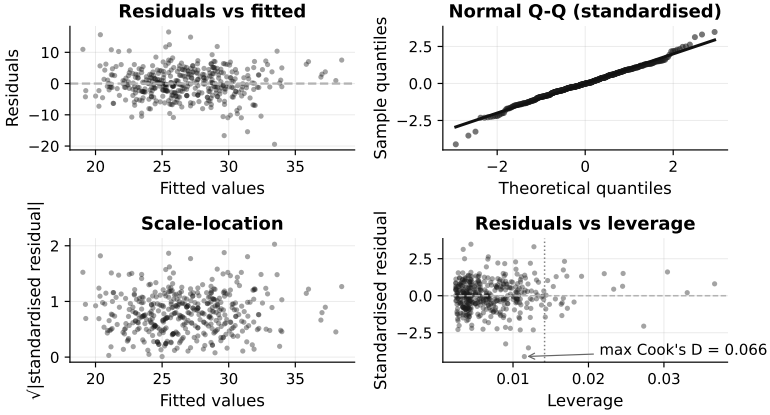
Each person answered once, and the data record no grouping — no household, class or site. That is all the design tells you here: a sample recruited through a few workplaces would be clustered without showing it.

Watch out

"Independence was met" with nothing examined is a claim with no evidence. Say how the data were collected, or say it was not examined.

Writing It Up

The Diagnostic Panel



`m9.residual_panel(model1)`: residuals against fitted values, the Q–Q plot, scale–location, and residuals against leverage. One model per figure. Collinearity is a table, not a picture.

Writing the Diagnostics Up

```
m9.report_diagnostics(
    model1,
    plots="Residuals scatter evenly about zero with no curve or funnel, "
          "and the Q-Q plot strays only in the tails",
    design="each respondent answered once, and the data record no grouping",
)
```

The paragraph

- Starts with your reading of the plots — it will not run without
- Each test with what it found, and only that
- Independence as judged from the design, or not examined

Written after fitting

A paragraph that "confirms" the assumptions before any model exists is a template, not a finding.

Break — 30 minutes: Variance, Collinearity and the Report

25 minutes on the analysis, then 5 to stretch.

In `notebook.ipynb`

- **Part 2 to the end**, then **Parts 3 and 4**

Then in `exercise.ipynb`

- Questions 3 and 4 — the sleep models, diagnosed and written up

Think about

Your diagnostic paragraph will not run without your reading of the plots. Write that sentence before you look at any test.

Then stop

Five minutes on your feet before the last section.

Where You Should Be: The Report

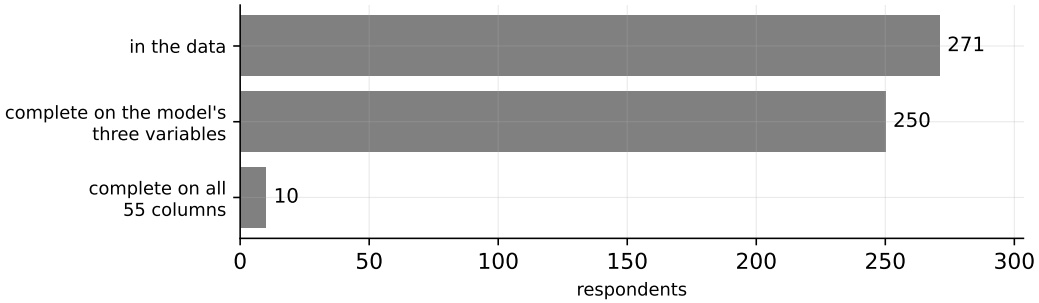
You should now have

- Breusch–Pagan, named, beside the residual plot
- VIFs for all three models, on the same rows
- Models 2 and 3 compared with nested tests
- A generated diagnostic paragraph, starting with your own sentence

The two that catch people

Calling an extra predictor a moderator. And writing “independence was met” about data whose collection was never described.

A Model Fitted on Ten People



Drop every row with a gap in any of the sleep study's 55 columns and 10 of 271 remain. A model of sleepiness on stress — still stored as text, so entered as 5 dummies — and weeknight sleep fitted to them reports $F(6, 3) = 106.2$ and 99.5% of the variance "explained". That is 7 coefficients, intercept included, for 10 people: the fit is the arithmetic, not the sleep. Selecting the model's three variables first keeps 250.

Conducting a Multiple Regression

1. **A question, and a reason for each predictor**
2. **Select the variables, then drop rows**, with the cost shown
3. **Look** at each variable and at every pair you will model
4. **Fit**, with any categories' references named
5. **Diagnose**: the residual plots first, then the tests, then the cases the model leans on
6. **Compare** nested models on the same rows
7. **Report** the model and its diagnostics, each generated, each saying what it does not establish

Reporting: Tables or Text

Several models: a table. One model: a paragraph. Never both for the same numbers.

A table carries

- Each coefficient with its interval
- R^2, F and N for each model
- One row per term; categories under their predictor

When to switch

When a paragraph has nearly as many numbers as words, it is a table. Follow your field's convention for which coefficients to show.

Two Conclusions You Must Not Draw

"The model met all its assumptions"

Where a check finds nothing, that is not the same as the assumption holding. Here two checks did find something — Shapiro–Wilk and the outlier test — and both were judged not to change the model. And no check examined independence.

"Control reduces stress"

The survey measured both once. Regression adjusts for what is in the model — here, social desirability — and for nothing else. It still describes an association.

Scope

You SHOULD be able to

- Compute and report a partial correlation, and say how it differs from a semi-partial one
- Fit on the rows the model needs, with the cost shown
- Read residual plots before tests, and say what each assumption is about
- Find outliers, leverage points and influential cases, and decide what to do with them
- Check constant variance and collinearity, and name the remedies
- Say how independence was judged
- Write a diagnostic paragraph that claims only what it found

That's all for this week